

Supplementary materials for "Predictive Modeling of Hospitalization from a Whole-Person Health Score with Incomplete Electronic Health Records Data: A Case Study"

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ADDITIONAL FIGURES

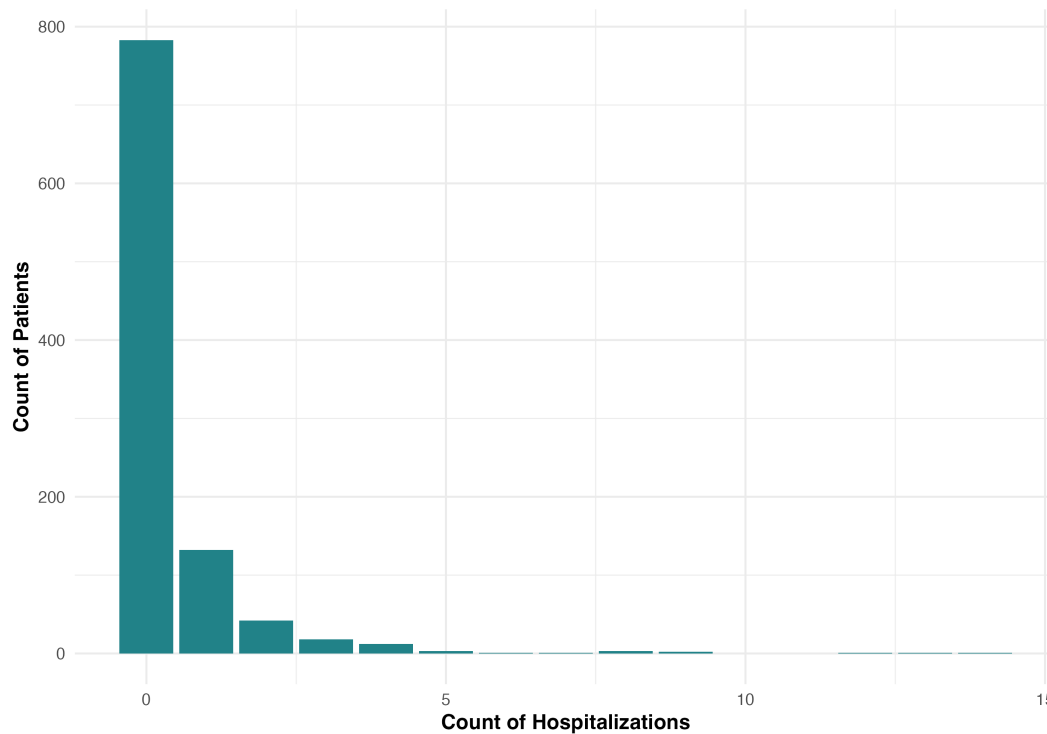


Figure S1. Distribution of number of hospitalizations in the 2-year study period across the $n = 1000$ patients in the sample from the electronic health records (EHR) at Atrium Health Wake Forest Baptist Hospital. Most patients were not hospitalized (79%), but some were admitted as many as 14 times during the 2-year study period.

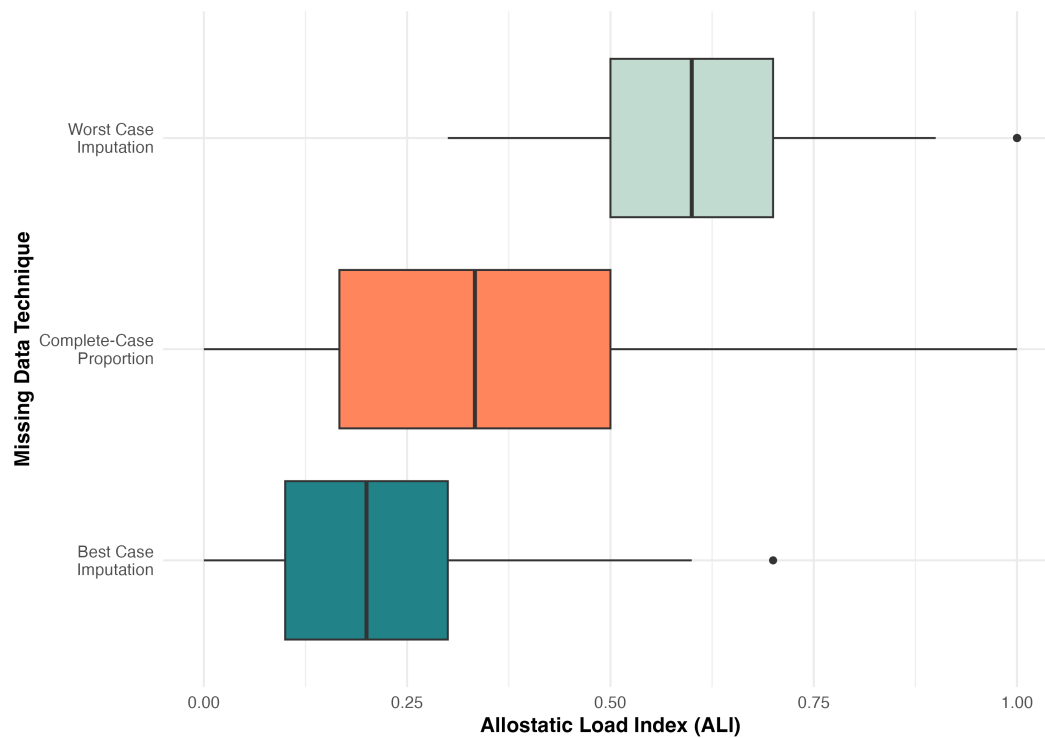


Figure S2. Distributions of the allostatic load index (ALI) for the $n = 1000$ patients in the sample from the electronic health records (EHR) at Atrium Health Wake Forest Baptist Hospital after using the complete-case proportion and best/worst case imputation missing data methods.

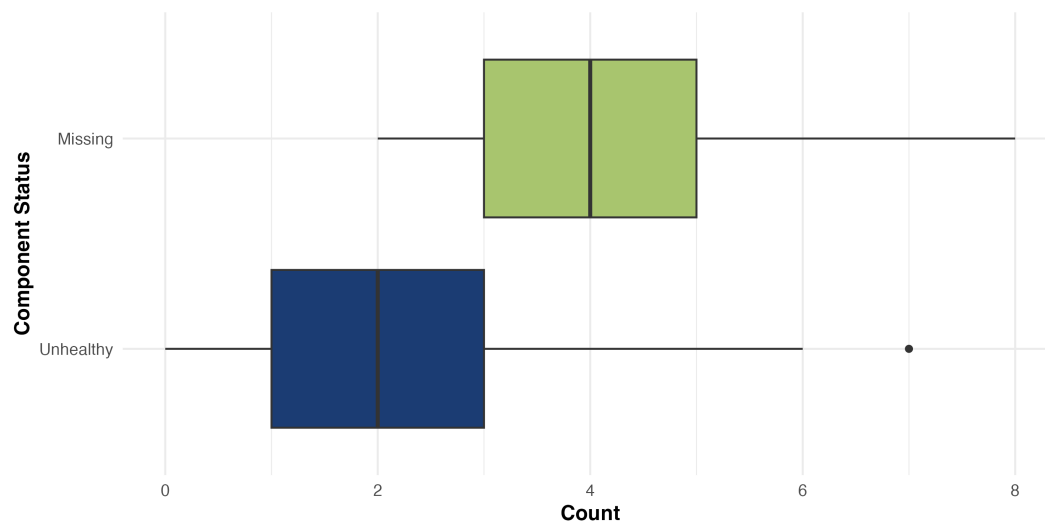


Figure S3. Distributions of the counts of unhealthy and missing allostatic load index (ALI) components for the $n = 1000$ patients in the sample from the electronic health records (EHR) at Atrium Health Wake Forest Baptist Hospital.

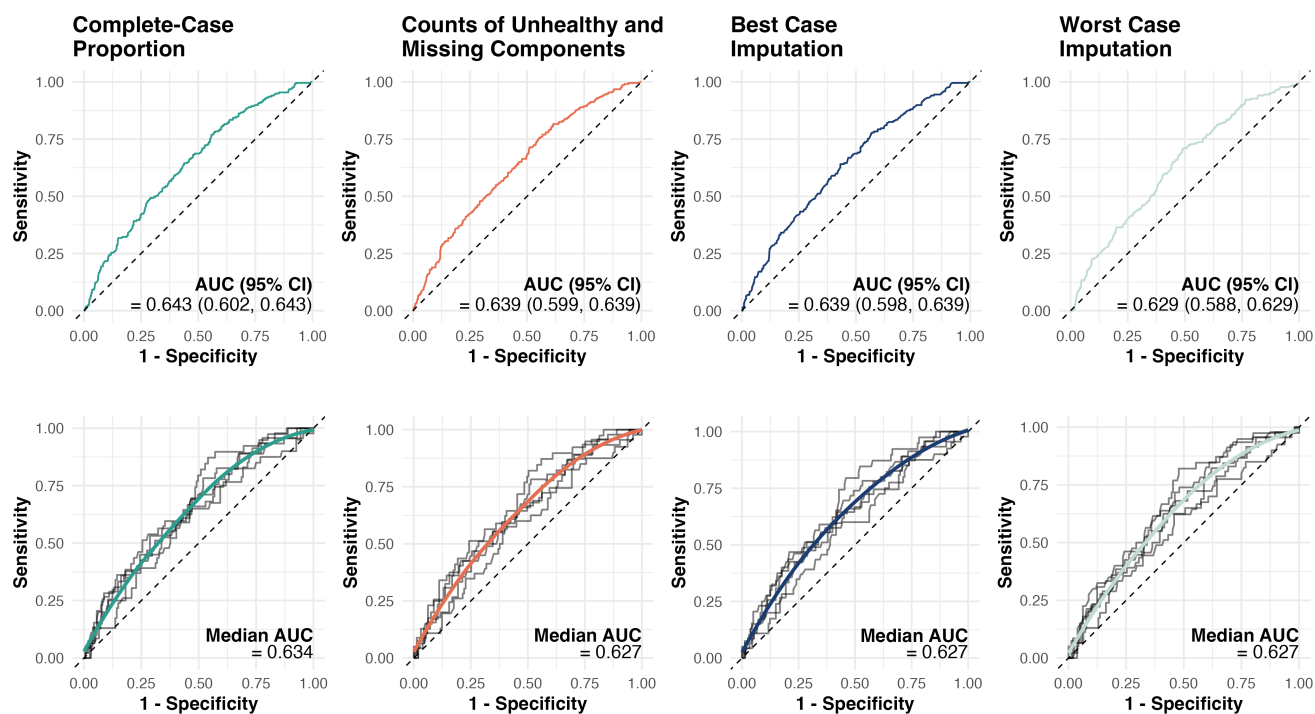


Figure S4. Receiver operating characteristic (ROC) curves for the four summary measure models using logistic regression based on the full sample data (top row) and 5-fold cross-validation (bottom row). For the full-sample models, the area under the ROC curve (AUC) and its 95% confidence interval (95% CI) are included. For the cross-validated models, the median AUC across the 5 folds is included.

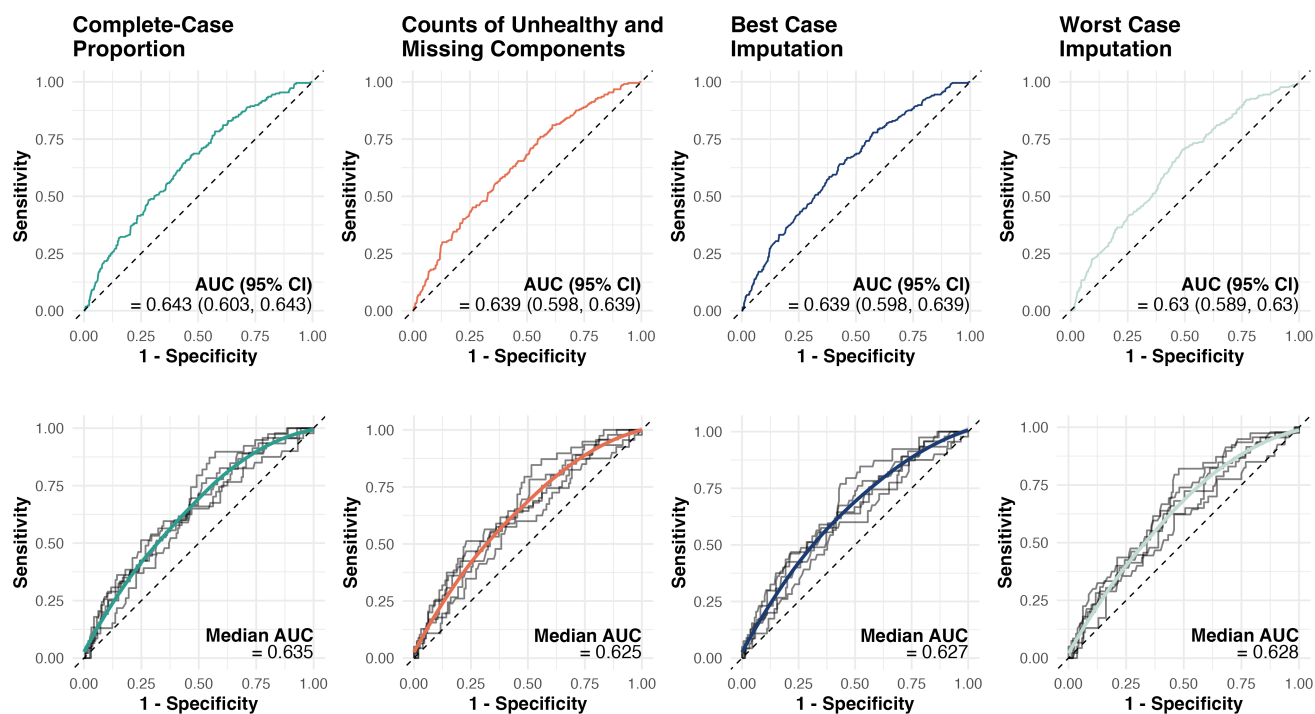


Figure S5. Receiver operating characteristic (ROC) curves for the four summary measure models using Poisson regression based on the full sample data (top row) and 5-fold cross-validation (bottom row). For the full-sample models, the area under the ROC curve (AUC) and its 95% confidence interval (95% CI) are included. For the cross-validated models, the median AUC across the 5 folds is included.

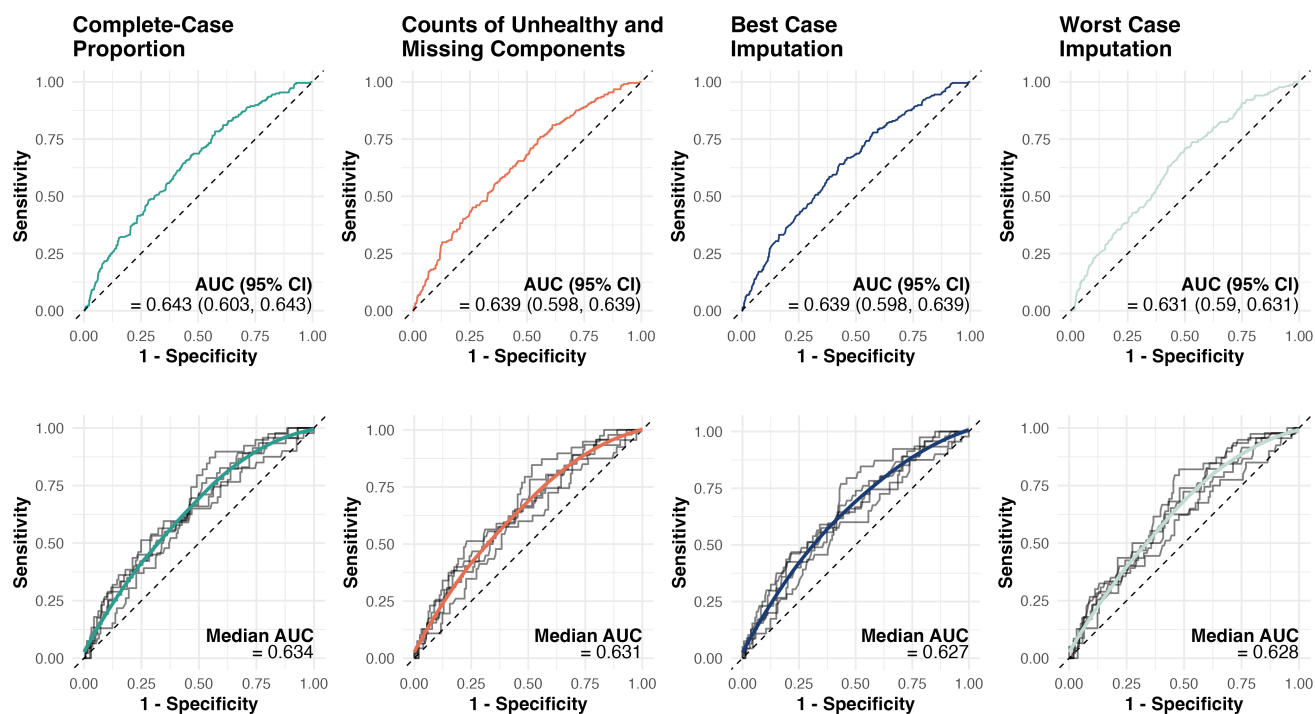


Figure S6. Receiver operating characteristic (ROC) curves for the four summary measure models using zero-inflated Poisson regression based on the full sample data (top row) and 5-fold cross-validation (bottom row). For the full-sample models, the area under the ROC curve (AUC) and its 95% confidence interval (95% CI) are included. For the cross-validated models, the median AUC across the 5 folds is included.

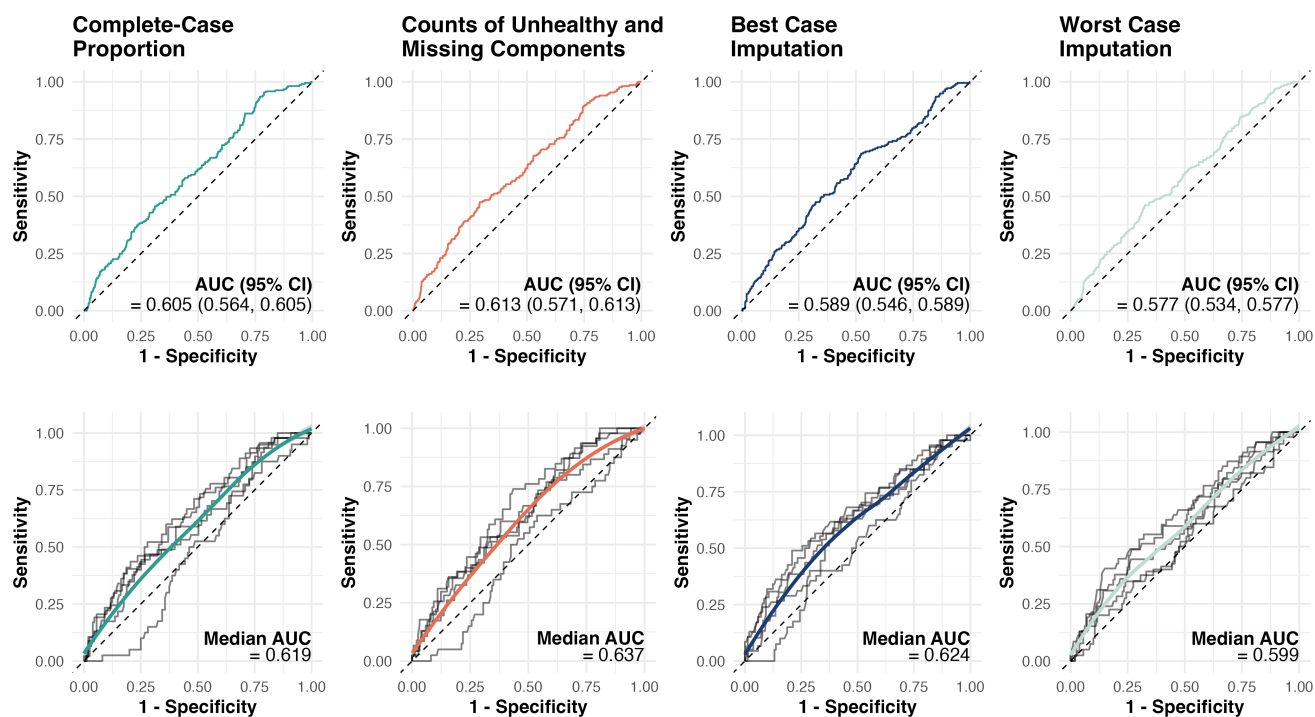


Figure S7. Receiver operating characteristic (ROC) curves for the four summary measure models using random forest classification based on the full sample data (top row) and 5-fold cross-validation (bottom row). For the full-sample models, the area under the ROC curve (AUC) and its 95% confidence interval (95% CI) are included. For the cross-validated models, the median AUC across the 5 folds is included.

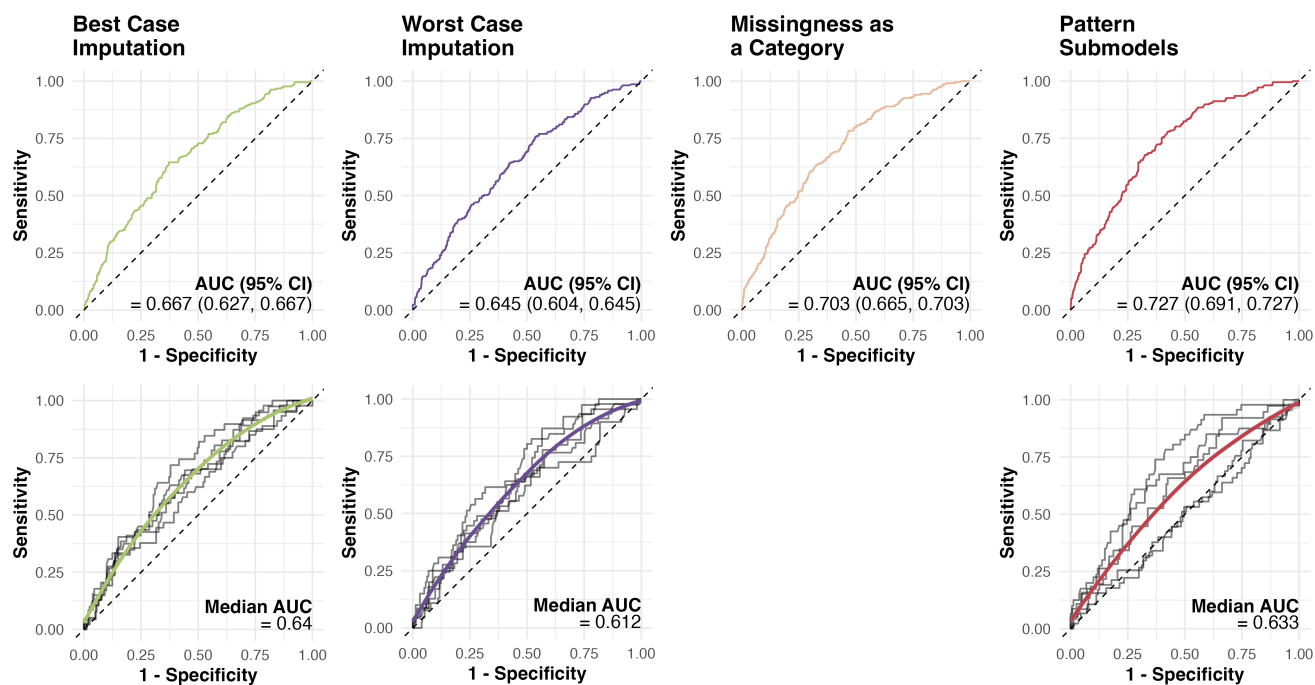


Figure S8. Receiver operating characteristic (ROC) curves for the four separate component models using logistic regression based on the full sample data (top row) and 5-fold cross-validation (bottom row). For the full-sample models, the area under the ROC curve (AUC) and its 95% confidence interval (95% CI) are included. For the cross-validated models, the median AUC across the 5 folds is included. The missingness as a category model could not be 5-fold cross-validated due to extremely rare categories, which would sometimes appear in the test but not train data.

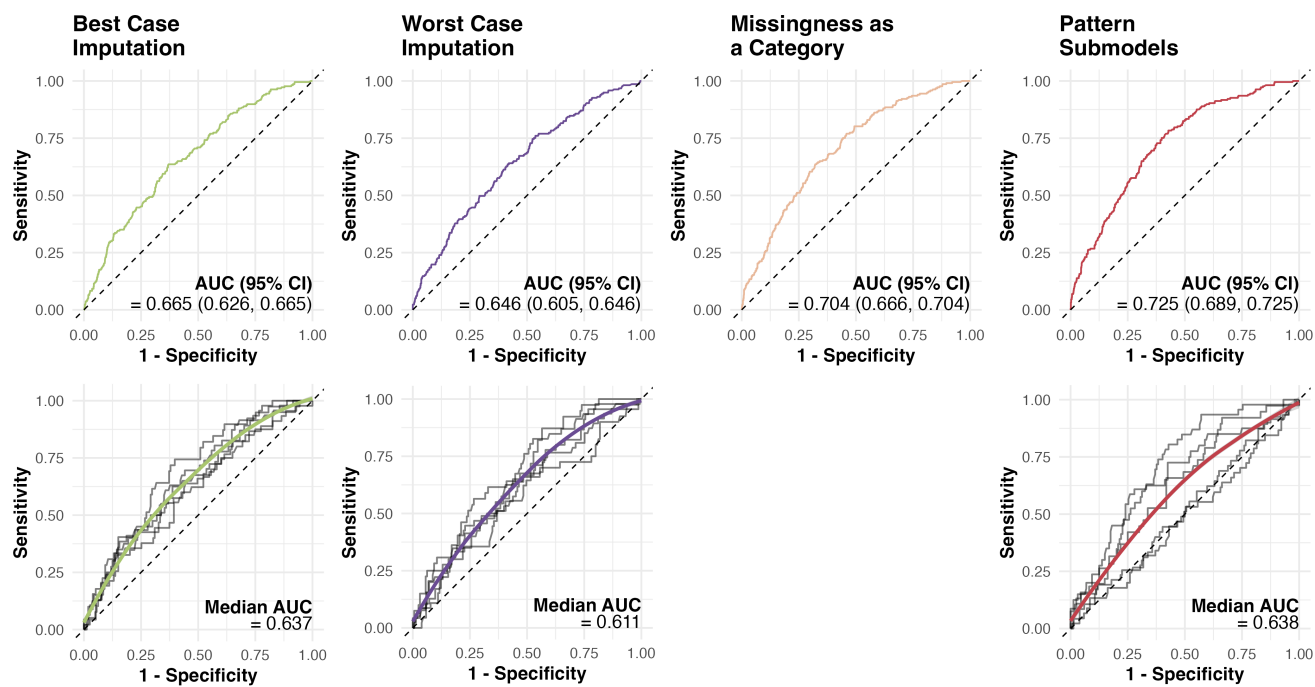


Figure S9. Receiver operating characteristic (ROC) curves for the four separate component models using Poisson regression based on the full sample data (top row) and 5-fold cross-validation (bottom row). For the full-sample models, the area under the ROC curve (AUC) and its 95% confidence interval (95% CI) are included. For the cross-validated models, the median AUC across the 5 folds is included. The missingness as a category model could not be 5-fold cross-validated due to extremely rare categories, which would sometimes appear in the test but not train data.

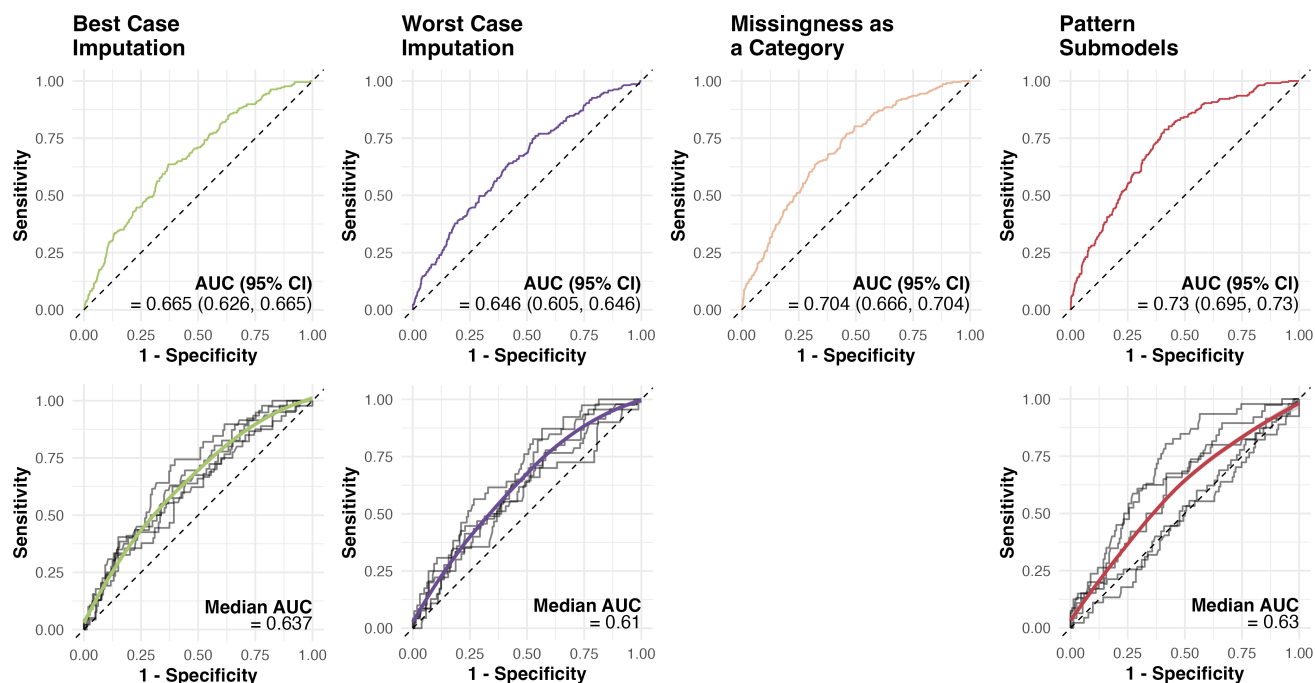


Figure S10. Receiver operating characteristic (ROC) curves for the four separate component models using zero-inflated Poisson regression based on the full sample data (top row) and 5-fold cross-validation (bottom row). For the full-sample models, the area under the ROC curve (AUC) and its 95% confidence interval (95% CI) are included. For the cross-validated models, the median AUC across the 5 folds is included. The missingness as a category model could not be 5-fold cross-validated due to extremely rare categories, which would sometimes appear in the test but not train data.

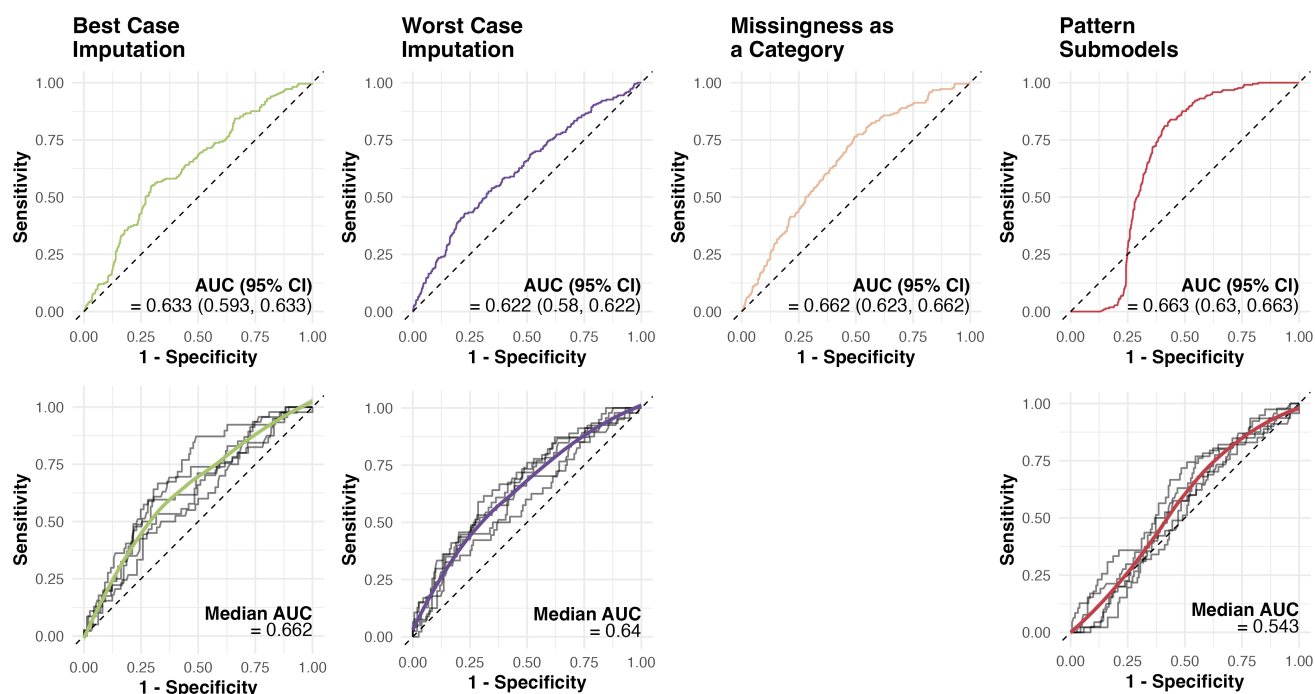


Figure S11. Receiver operating characteristic (ROC) curves for the four separate component models using random forest classification based on the full sample data (top row) and 5-fold cross-validation (bottom row). For the full-sample models, the area under the ROC curve (AUC) and its 95% confidence interval (95% CI) are included. For the cross-validated models, the median AUC across the 5 folds is included. The missingness as a category model could not be 5-fold cross-validated due to extremely rare categories, which would sometimes appear in the test but not train data.

STATISTICAL MODELING OF HOSPITALIZATIONS AS A COUNT

Let Y_C be a count version of the outcome ($Y_C \in \{0, 1, \dots\}$), representing the number of hospital admissions a patient had in the 2-year study period. We want to predict Y_C from the allostatic load index (ALI) X , controlling for additional covariates (age and binary biological sex), denoted together as $\mathbf{Z} = (Z_1, Z_2)$.

Poisson Regression

For the count case, we began by considering a *Poisson regression model*:

$$\log\{E_{\beta}(Y_C | X, \mathbf{Z})\} = \beta_0 + \beta_1 X + \beta_2^{\top} \mathbf{Z}, \quad (\text{S.1})$$

where the left-hand side of (S.1) denotes a patient's conditional log-average number of hospitalizations. and $\beta = (\beta_0, \beta_1, \beta_2)^{\top}$. In R, the built-in `glm` function can fit Poisson models.¹ While Poisson regression is a very common approach to modeling count outcomes, we faced two potential challenges in applying it to our sample data.

Challenge 1: Overdispersion

First, the Poisson model is that it assumes equidispersion, such that the conditional variance of the outcome $V_{\beta}(Y_C | X, \mathbf{Z})$ is equal to its conditional mean $E_{\beta}(Y_C | X, \mathbf{Z})$. There are various ways to check this assumption, including hypothesis tests as with the `dispersiontest` function in the *AER* R package.² When the assumption is violated, the standard error estimates from the fitted Poisson regression model will be too small, leading to issues in downstream inference (e.g., inflated test statistics, too-small p-values, and lowered statistical power). However, since we are using the model for prediction, switching to a model like negative binomial that accommodates overdispersion, when $V_{\beta}(Y_C | X, \mathbf{Z}) > E_{\beta}(Y_C | X, \mathbf{Z})$, is not expected to offer any improvements.³

Challenge 2: Zero Inflation

Second, zero inflation in count data occurs when a dataset contains an excessive number of zeros (relative to what the Poisson distribution expects) and is a violation of the assumptions of standard models. In this sample, 783 patients (78.3%) had no hospitalizations during the 2-year study period. Theoretically, these 783 patients represent two distinct populations: those who would never be hospitalized (e.g., due to access barriers) versus those who might have been hospitalized but simply were not during this 2-year period. We did consider an alternative model for Y_C to allow for zero inflation.

Zero-Inflated Poisson Regression

The zero-inflated Poisson (ZIP) regression model assumes that, conditional on X and $\mathbf{Z}$, the observed outcome Y_C arises from a mixture of a point mass at zero, representing patients with no risk of hospitalizations during the 2-year study period, and a Poisson process, representing the hospitalization counts among patients actually at risk. In other words, the ZIP model allows for differentiation between the zero counts in the model from patients who were *structural zeros*, meaning they had no risk of being hospitalized, and patients who were *sampling zeros*, meaning they could have been hospitalized but were not during this particular time period.

Let π be the probability that a patient belongs to the structural zero group and $\mu = E_{\beta}(Y_C | X, \mathbf{Z})$ be the conditional average count of hospitalizations, given that the patient was at risk for hospitalization. Then, the ZIP model is a mixture based on the following conditional probability function:

$$\Pr_{\beta, \pi}(Y = y | X, \mathbf{Z}) = \begin{cases} \pi + (1 - \pi)\Pr_{\beta}(Y = 0 | X, \mathbf{Z}) & y = 0 \\ (1 - \pi)\Pr_{\beta, \pi}(Y = y | X, \mathbf{Z}) & y = 1, 2, \dots \end{cases} \quad (\text{S.2})$$

where $\Pr_{\beta}(Y = y | X, \mathbf{Z})$ is the Poisson probability function governing the model in (S.1). The probability parameter π describes the distribution of the binary group membership variable G , which is defined such that $G = 1$ if the patient belongs to the structural zero group and $= 0$ otherwise. Additional covariates can be used to try to “guess” what G would be for each patient. Still, this variable is latent, and statistical methods are needed to estimate $\hat{\pi}$ and the Poisson coefficients $\hat{\beta}$, where the latter are now interpreted among the subset of patients who were at risk for hospitalization. In R, the `zeroinfl` function from the *pscl* package can fit a variety of zero-inflated models, including Poisson.⁴

MISSING DATA TECHNIQUES WHEN MODELING WITH COMPONENTS SEPARATELY

Best/worst case imputation

We can also replace missing components with the best/worst case scenario and then keep them as separate predictors in the models, rather than aggregating them into proportions $\check{X}^B$ or $\check{X}^W$ from the main text. In other words, let

$$\check{S}_j^B = \begin{cases} S_j & \text{if } R_j = 1 \\ 1 & \text{if } R_j = 0 \end{cases} \quad \text{and} \quad \check{S}_j^W = \begin{cases} S_j & \text{if } R_j = 1 \\ 0 & \text{if } R_j = 0 \end{cases} \quad (\text{S.3})$$

denote the imputed j th component under best and worst case, respectively ($j \in \{1, \dots, 10\}$). Then, for a chosen scenario (best or worst case), we fit the desired model using the 10 imputed components in place of X and any other covariates $\mathbf{Z}$. For example, the logistic regression model using best-case imputation (separate components) is

$$\log \left\{ \frac{\Pr_{\check{\beta}^B}(Y = 1 \mid \check{\mathbf{S}}^B, \mathbf{Z})}{1 - \Pr_{\check{\beta}^B}(Y = 1 \mid \check{\mathbf{S}}^B, \mathbf{Z})} \right\} = \check{\beta}_0^B + \sum_{j=1}^{10} \check{\beta}_{1j}^B \check{S}_j^B + \check{\beta}_2^{B\top} \mathbf{Z}, \quad (\text{S.4})$$

where $\check{\mathbf{S}}^B$ is the vector of best-case components imputed as in the first equality of (S.3).

In our application to electronic health records (EHR) data for $n = 1000$ patients routinely engaging in care at Atrium Health Wake Forest Baptist Hospital, there were some components that (i) were heavily missing and (ii) had only healthy values that for the few that were non-missing. Thus, after imputing the many missing values with the best case scenario (healthy), these components were constant across all patients in the sample and had to be excluded from models. In particular, creatinine clearance and homocysteine were constant after best-case imputation, and were excluded from the separate component models accordingly.

Connection between missingness as a category and counts of unhealthy and missing components

As mentioned in the main text, the missingness as a category approach could be seen as a generalization of the counts of unhealthy and missing components approach, wherein the individual components can have different coefficients. Beginning with the right-hand side of (4), notice that

$$\begin{aligned} \beta_0^C + \beta_1^C C_S + \beta_2^C C_M + \beta_3^{C\top} \mathbf{Z} &= \beta_0^C + \beta_1^C \sum_{j=1}^{10} S_j + \beta_2^C \sum_{j=1}^{10} M_j + \beta_3^{C\top} \mathbf{Z} \\ &= \beta_0^C + \beta_1^C (S_1 + \dots + S_{10}) + \beta_2^C (M_1 + \dots + M_{10}) + \beta_3^{C\top} \mathbf{Z}, \end{aligned}$$

which can be interpreted as a model where missingness was treated as a category *but* the coefficients β_1 and β_2 controlling the adjusted associations between unhealthy and missing, respectively, and hospitalization were assumed to be the same for all ALI components. Using notation for missingness as a category from (8), the counts of unhealthy and missing components asserts that $\check{\beta}_{1j} = \beta_1$ and $\check{\beta}_{2j} = \beta_2$ for all components j ($j \in \{1, \dots, 10\}$).

Pattern submodels

As proposed by Fletcher Mercaldo and Blume (2020), the pattern submodels approach was implemented as follows.⁵ First, we identified the 18 unique missing data patterns present in our sample of $n = 1000$ patients (illustrated in Figure 2). Then, going through these missing data patterns one at a time, we subset to the patients within each subgroup. For example, the first pattern was missing creatinine clearance, C-reactive protein, and homocysteine, and there were 392 patients with this pattern in their EHR data. This pattern's submodel included the other seven non-missing ALI components $\mathbf{S}^1$ (systolic/diastolic blood pressure, body mass index, triglycerides, total cholesterol, hemoglobin A1c, serum albumin) and additional covariates $\mathbf{Z}$ (age and sex). As with *missingness as a category*, ALI components could be constant within some patterns (e.g., all patients had healthy values), in which case those components were excluded from the submodel accordingly.

As displayed in Figure 2 of the main text, there were also some patterns seen for very few patients. Therefore, these patterns had insufficient data to fit their own submodel in the way described previously. Instead, we tried to fit what Fletcher Mercaldo and Blume (2020)⁵ called a *complete case submodel (CCS)*. Consider for example the eighth pattern, where 5 patients were missing creatinine clearance and homocysteine. This group's CCS would

use have used *all* patients out of the original 1000 that had non-missing values for the other eight ALI components (systolic/diastolic blood pressure, body mass index, triglycerides, total cholesterol, hemoglobin A1c, serum albumin, C-reactive protein), rather than just the 5 in the subgroup.

Still, in our sample, there were patterns where even the CCS subset was insufficient. In these cases, we “nested” the missing data pattern within another that was better-supported in the sample and missing the same components (and then some). In other words, we assumed that the same submodel from a better-supported missing data pattern held for this one, as well. For example, the eighth pattern (described previously), was nested within the first pattern, which was missing the same two components *plus* C-reactive protein. However, this stability, while necessary, comes at a cost. Nesting the eighth pattern meant that the 5 patients in that subgroup did not actually have their non-missing C-reactive protein used for their predictions. For the R code implementation, see www.github.com/sarahlotspeich/missALI.

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